

Co-designing Resources for Ethics Education in HCI

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ABSTRACT

Due to the evolving nature of technology and its impact on individuals, communities and society, practitioners and designers in Human-Computer Interaction (HCI) are expected to consider ethics in their work. This role has inspired the development of a number of resources for practice, such as tools, frameworks and methods to tackle ethical issues in HCI. But these suffer from low adoption rate potentially because they are not yet part of the standard body of knowledge. To mitigate the issue, we argue that there is an urgent need for ethics education in HCI. Beyond defining ethics, an ethics curriculum must enable practitioners to reflect and allow consideration of intended and unintended consequences of the technologies they create from the ground up, rather than as a fix or an afterthought. In this co-design workshop, we aim to build upon existing practices and knowledge of ethics in HCI and work with the CHI community to enrich ethics curriculum. We will scaffold our collective understandings of the existing resources and create guidelines that support interactive educational experiences to support HCI ethics curriculum.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interactive systems and tools**;

KEYWORDS

Applied Ethics, HCI Education, Co-design

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1 INTRODUCTION

With new technologies seeping into every aspect of our personal, social and public life, a myriad of ethical issues and concerns have come to fore. However, many of these technologies lack foresight or deliberately forego those ethical concerns. Examples abound. For instance, a mSpy mobile app aimed at stopping cyberbullying by allowing parents to monitor their children's activities (e.g., messages, calls, location and social media), turned into a tool for cyberstalking itself. The app was subsequently removed from Google's Play store in 2013 [21]. In the wake of the COVID-19 pandemic, a number of mobile and QR code-based solutions were developed which sparked discussions around privacy violations globally [11]. It was soon made clear that many citizens willingly participate in risky online behaviors, such as sharing their GPS location, personal ID, phone numbers and addresses without verifying the authority collecting

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the data once the QR code is scanned, many users do not read the privacy policy or confirm SSL certificate of the website. Hoffman et al. examined alternative to such examples, using Zwaai app in the Netherlands [12]. They highlight the potentials for design decisions to surface ethical-political seams to make the complexities of privacy issues visible to users (contrary to seamless design), through different design features such as customizing configuration.

Where we fall on the ethical spectrum as Human-Computer Interaction (HCI) researchers and practitioners often depends on the interplay of our design decisions and awareness of ethical pitfalls as well as our ability to identify and mitigate the ethical issues before they arise. When is the best time to inform the design decisions? How do we raise awareness? What tools and methods should we use to identify and address urgent ethical questions? In this workshop we trace these questions to the origins of practice: the HCI education. Unless ethics is integrated in every aspect of design process and educational curriculum, it is bound to be an afterthought and thus inadequate in identifying and addressing ethical issues. With this premise, we investigate and co-design ways through which ethics can be fostered within the HCI curriculum.

The impact of interaction design and HCI decisions on emerging fields such as health, finance, education, politics is undeniable. The HCI community has a crucial influence on the behavior, emotion, psychology and wellbeing of people with clear ethical implications. In cases such as mSpy referenced earlier, the failure to predict potential unintended outcomes could lead to irreversible harms. Designers bear the responsibility of predicting and preventing such outcomes, which necessitates taking an active role prior to or during the development.

In recent years, we have seen a number of calls to action on ethics within the HCI community. Multiple workshops at CHI 2020 explored ethics in a range of topics such as cognitive bias in computing systems [2], ethics of user experience design [4], responsible AI [16] and ethical implications of mixed realities [10]. A series of publications brought to light the dark patterns in user experience (UX) [8] calling for ethical UX pedagogy [6]. Others draw attention to the political use of “unintended consequences” within HCI and technology discourses to illustrate the vital but often overlooked role of ethics and responsibility in design [17]. It is clear that the complex nature of interactive technologies mandates HCI researchers and practitioners to play an additional role of being ethicists and navigate potential consequences of their creations with intent and speculate responsible alternatives.

How would the HCI practice be different if the designers were educated in ethics? UX designers are tasked with considering how their design decisions provide enjoyable emotional and behavioral outcomes [11]. Could this be extended to ethical consequences, to provide scope and tools to challenge initial design assumptions, while also empowering reflection on our own ethical grounds [25]? Developing a fine-tuned perspective of ethics requires an ability to examine the intended as well as unintended consequences of design for individuals, communities and society. However, education in ethics is still lacking from HCI and design curriculum to a large extent [7]. In this workshop we explore and co-create ways of addressing this gap.

The addition of ethics courses and topics to existing curriculum has been growing in engineering and computer science faculties

in many universities [3] to introduce theory and enable conversations around ethics adjacent to topics such as values, responsible innovation and justice [24]. However, there is still a need for building better guidelines on how these curricula can be designed and situated, and how they can be accessible to UX practitioners and researchers in addition to students. We aim to bring together practitioners, educators, researchers and students to capture their ethical perspectives, and identify how we can co-design taxonomies to empower discussion around ethical education that supports responsible decision making, critique of existing systems, and reflection on ethics approaches in HCI practice.

1.1 Limitations of Ethics Heuristics

Public outcries over unintended consequences of commercial technologies often trigger discussions around the need for an ethics oversight. This, while necessary, may be considered a fix, an afterthought. As an example, the ride sharing service Uber updated their system to make visible the details of fare and distance to provide drivers with more control. This fix came after complaints about how the system persuaded drivers to stay on the job for longer hours because encouraging push notifications reminded them about how close they are to achieving their goal, alerting them of the upcoming fares even before a ride has been completed, or not revealing the distance and location of rides in advance, all designed to increase the bookings [23]. Yet, other ethical issues raised by the introduction of such systems, such as those of relational justice [15], remain unexamined.

The need for education in ethics has been renewed in recent years [6]. The desire to address ethics in emerging technologies have resulted in the development of heuristic tools and checklists such as ethical technology assessments [14], Ethical OS toolkit [28], and Digital Impact toolkit [29], among others. Kiran et al. [14] noted, however, that while this is a step towards the right direction, heuristic tools are not well integrated into the process of design and a more constructive approach is needed. Beyond checklists, responsible design frameworks and principles, for instance for AI [20], address that gap through a process-driven approach to ethics with great appreciation for human wellbeing. Design methods such as design fiction, co-design were additionally proposed suitable to explore ethics in HCI [27].

We take a step further and argue that the ability to integrate any tool or framework into practice should be cultivated through the HCI curriculum. Ethical tools and frameworks alone do not produce ethical outcomes. It is up to the practitioners to put those frameworks to work as they confront inherently uncertain situations of practice, for instance in negotiating the relationship between design and values [13]. This is our impetus for inviting the CHI community to examine how the ethics resources (e.g., frameworks, themes and methods) may be integrated in the future HCI curriculum.

1.2 Renewing the Call for Ethics Education in HCI

A number of existing methodologies are consistently understood as core to HCI curriculum, for instance value sensitive design [5] [13], responsible innovation [1], collaborative design [19], design for wellbeing [18]. Adopting these methodologies add new utility for

HCI education, however a grounding of ethics into practice is not guaranteed. We aim to synthesize existing approaches to establish a guideline for best practices on how ethics resources can align with the existing dominant pedagogical direction.

In this workshop, we invite participants to collaborate towards a shared understanding of ethics curriculum in HCI and the challenges and opportunities it affords. In doing so, we view ethics not as a set of principles for an idealized world but as a means to educate accountable HCI researchers, designers and developers who are self-aware of the impact of their decisions on others, who are capable of making informed decisions about uncertain technology consequences and are agents of change who willingly shoulder the responsibility of cultivating justice and fairness by reflecting on their own practices.

Because new technology capabilities such as mixed realities and Artificial Intelligence (AI) are constantly evolving, we see new impacts emerging over time. As such, ethics education, needs to be responsive to these changes [26]. We aim to discuss how an adaptive educational experience can provide an introduction to ethics resources, while also generating an ability in students to reflect on their own ethical views. A desirable outcome would be to empower individuals to build the core ethical principles in their own organizations that can be readily integrated into their practice.

1.3 Co-Designing the Resources of Ethics Education in HCI

In this workshop, we will facilitate a discussion to capture the current landscape of ethics resources used in education and practice. Through co-design activities, we then generate new ways of educating the future HCI generation. Teaching ethics in HCI means working in an environment that is ambiguous, requiring students to experience multiple perspectives and traverse through uncertainty. As such, we argue there is a requisite to use purpose built artefacts and folios through participatory, relational and sense-making activities to accelerate the learning experience and help individuals embrace uncertainty [9], especially in a field like ethics. In this workshop, we take the first step towards developing a guideline for creating such resources.

These resources will inform the building blocks of ethics curriculum in HCI. We will use co-design activities in the workshops because they are effective for reflecting and recognizing diverse perspectives [22]. Through collective exploration, we hope to capture the existing and emergent knowledge of ethics in action within the CHI community. The outcome of the workshop will help participants to reflect on their understanding of ethics. It will then allow us to develop our own situated ethics education resources.

2 KEY QUESTIONS

This one-day workshop aims to understand and make visible the current use of ethics in HCI practice and education. Through co-design, we will then explore guideline for establishing resources for HCI education. Through workshop activities, participants will engage with the following questions:

- What does ethics mean for HCI?
- What ethics resources are currently identified as effective in HCI research and practice?

- What are the best ethics resources (methods, tools, techniques) used in HCI curriculum?
- What are the opportunities and challenges for improving ethics education in HCI curriculum?

3 WORKSHOP GOALS

The workshop will focus on inter-disciplinary discussion between participants who identify as practitioners, researchers, educator, students and industry partners in the field of HCI with particular interest in ethics. We want to identify ways we may help future students afford new perspectives that may not be accessible through the familiar lenses, because interactive technologies are constantly evolving. Our HCI education must prepare future practitioners for negotiating the ethics of technologies that are not created yet. During the workshop we will accomplish the following goals:

- Achieve a common and community-based understanding of ethics curriculum in HCI
- Collate different interactive and experiential modes for teaching ethics in HCI
- Identify and reach consensus on future directions for improving ethics curriculum in HCI
- Plan for actions beyond the workshop

4 ORGANIZERS

Ajit G. Pillai is a researcher at Design Lab in The University of Sydney. He regularly conducts research on the ethics of emerging technologies in HCI in industries such as healthcare. He is interested in the future of ethics education in HCI and is exploring interactive resources to augment learning experiences in effective and interactive ways.

A. Baki Kocaballi is lecturer at the School of Computer Science, The University of Technology Sydney. His research situated within the intersection of interaction design, AI, and digital health with a specific focus on multimodal and conversational technologies. He is interested in employing relational, responsible, and generative design approaches to understand and re-imagine human-AI relationships.

Tuck Wah Leong specializes in Participatory Design (PD). In PD, participation is not only about inclusion but also about ethics, politics, democracy and empowerment. Since PD is intrinsically focused on co-designing with people's values, Tuck has worked on methods and processes to support co-discovery and negotiation of individual and collective values with stakeholders and participants.

Rafael A. Calvo is Professor at the Dyson School of Design Engineering, Imperial College London. He is also co-lead at the Leverhulme Centre for the Future of Intelligence (Imperial spoke). He focuses on the design of systems that support wellbeing in areas of mental health, medicine and education, and on the ethical challenges raised by new technologies.

Nassim Parvin is an Associate Professor of Digital Media in the School of Literature, Media, and Communication at the Georgia Institute of Technology where she directs the Design and Social Justice Studio. Dr. Parvin also serves as the co-lead editor of Catalyst: Feminism, Theory, and Technoscience. Her research explores the ethical and political dimensions of design and technology.

Katie Shilton is an Associate Professor in the College of Information Studies at the University of Maryland, College Park. Her research explores ethics and policy for the design of information technologies. She is the PI of the PERVADE project, a multi-campus collaboration focused on big data research ethics.

Jenny Waycott is a Senior Lecturer in the School of Computing and Information Systems at The University of Melbourne. Her current research investigates ethical issues in the use of technology in aged care and examines how new technologies, such as virtual reality, can be designed and used to provide enrichment in later life.

Casey Fiesler is an assistant professor in information science at University of Colorado Boulder. Her research largely focuses on technology ethics and governance, including research ethics and ethics education for computer science. As part of her CS education work, she is part of the Responsible Computer Science Challenge.

John C. Havens is Executive Director of The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems and The Council on Extended Intelligence, a joint program between The IEEE Standards Association and MIT. He is the author of *Heartificial Intelligence: Embracing our Humanity to Maximize Machines*.

Naseem Ahmadpour is Senior Lecturer (Assistant professor) and the director of Affective Interactions lab, The University of Sydney. She conducts research at the cross-section of HCI, design, and psychology, and regularly publishes in HCI journals and conferences. She is interested in researching emerging technologies that support human flourishing through reflection and affective experiences.

5 WEBSITE

The workshop website will be hosted at <https://sites.google.com/view/codesignethics>. We will post the call for participation, position paper deadlines, detailed activities about the workshop and accepted position papers. Once the workshop has been completed, we will update the webpage to reflect on the outcomes for the CHI community to access. We will continue using the website as a forum for discussion and community building in the future.

6 PRE-WORKSHOP PLANS

The call for participation will be advertised through personal contacts, professional networks, social media and mailing lists. Furthermore, we will directly contact researchers and practitioners who are likely to be interested in the workshop. Participants are invited to submit position papers to the workshop in order to express their interest. Those who are accepted will be invited to present their work at the workshop and partake in workshop activities.

7 WORKSHOP STRUCTURE

This workshop will be conducted online via Zoom and Miro

Introduction and position paper presentations: The workshop will begin with an introduction to the topic by the organizers. Next, participants will briefly present their proposals to give the group an opportunity to become familiar with cutting edge approaches to ethics education and practices.

Reflective exercise: A facilitated reflective activity will follow to capture the current landscape of ethics resources used by participants in research, teaching and practice. This will help participants to additionally gain an understanding of their own ethical practices. The exercise involves working in groups and using a set of visual templates and post-it notes provided by the organizers. Participants will start by sharing personal stories, then identifying challenges and opportunities based on their experiences. A group discussion will be then facilitated with the aim of establishing shared understandings and best practices around the needs and challenges for curriculum reform through an ethical lens.

Building the future: Finally, co-design activities on Miro will help participants to speculate how future ethics curriculum and resources can supplement HCI education. This is in form of a roadmap that extends from 'now' to 'future'. Again, this activity is facilitated through templates provided to participants who will use post-it notes to share their thoughts (on Miro).

Final discussion: Finally, those who worked in groups will present their speculated futures. All participants will engage in a discussion around the opportunities for scaffolding the ethics curriculum in HCI and reflect on how a collective future may be built. Participants will be then invited to contribute to a statement of mission to achieve this future.

Wrap up: The workshop will end with a self-reflection at this point as participants will identify how they will use the insights from the workshop in their practice.

7.1 Workshop Schedule

09:00-09:30 Welcome and introductions

09:30-10:30 Position paper presentation by workshop participants

10:30-10:35 Break

10:35-11:00 Reflection exercise

11:00-12:00 Building the future: speculating a future ethics curriculum

12:00-12:05 Lunch Break

12:05-12:30 Final discussion

12:30-13:00 Wrap up

8 POST-WORKSHOP PLANS

Workshop outputs, including materials created during the co-design session will be shared with the wider HCI community through our website and subsequent publications. We will also discuss plans for future outputs with workshop attendees on the day, as we aim to build a community, continue the conversation and explore the potentials for collaborations.

9 CALL FOR PARTICIPATION

This one-day workshop at CHI 2021 invites practitioners, researchers, educators and students in HCI, co-design, user experience design, design education, ethical design, and emerging technologies to explore the need for ethics education in HCI. Participants are invited to co-create a vision of future ethics curriculum in HCI programs and professional developments. This workshop aims to generate an understanding of the current practices in ethics education in HCI, identify ways through which ethics education in

HCI can be enhanced through interactive experiences and forge directions for future developments. The outcomes will include a better understanding of the current state of ethics curriculum in HCI, future directions as well as how we might co-create future resources for teaching ethics in higher education.

Participants in the workshop will have an opportunity to present their views and works in this space, reflect on their own teaching, practices and understandings of ethics, and engage with other practices in the community. Participants are encouraged to submit position papers to express their interest in the workshop, describing one or more of the following: case studies, literature review, curriculum review, examples of projects relevant to ethics, ethics education, responsible HCI education or practice, possible examples of ethical pitfalls and challenges of working with ethics resources (e.g., frameworks, themes and methods), new and emerging resources to inform ethics HCI education.

Position paper should be 2-4 pages long (excluding references) in the ACM SIGCHI Master Article template. Submissions will be juried by the organizers based on originality and relevance. Submissions and questions should be emailed to . At least one author of each accepted paper must attend the workshop.

Call for participation: 15 Dec 2020, 12pm PT

Position papers due: 21 February, 12pm PT

Notifications due: 28 February

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